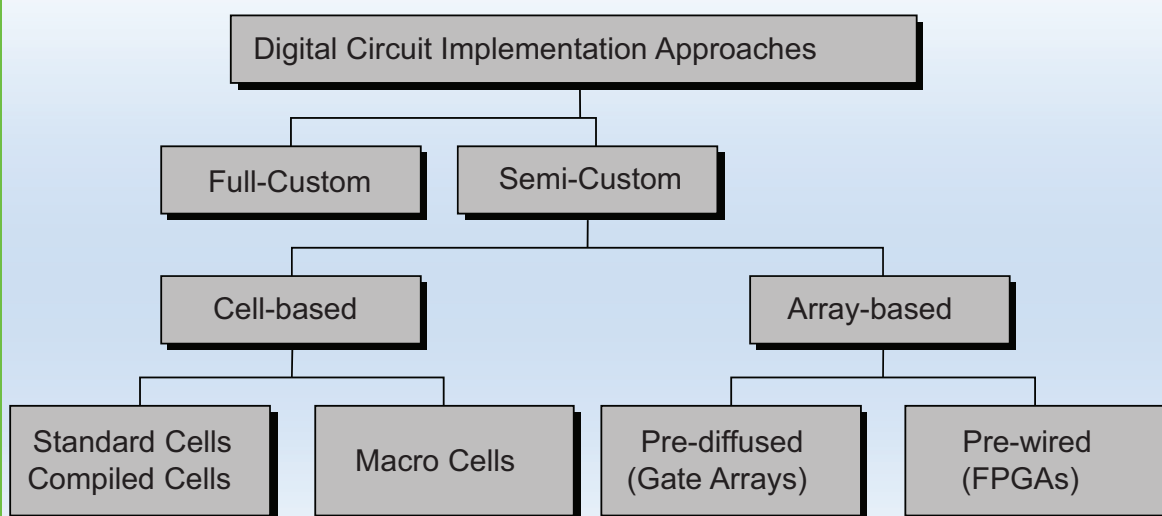


Design Methodology

Part I Full-Custom Design
Semi-Custom Design: Cell-based

Part II Semi-Custom Design: Array-based

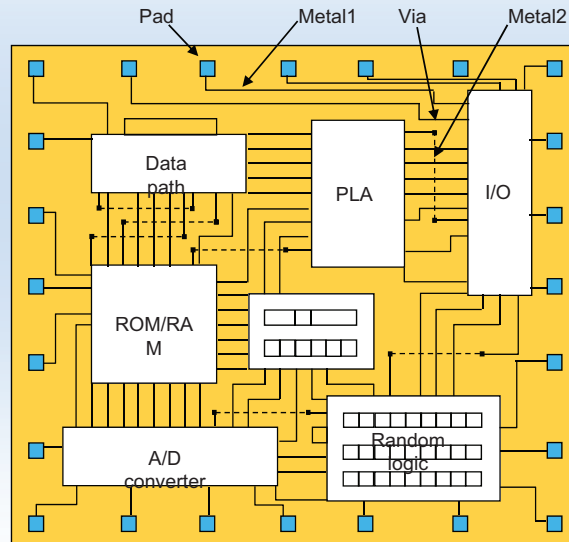
Circuit Implementation Choices



[fig. from: Rabaey et al, Digital Integrated Circuits, 2nd Edition]

Full Custom Design Approach

- This method allows full control over each wire and transistor width to make the fastest and smallest implementation of a design.



Full Custom Design Approach

The designer has many degrees of freedom.

- Free to optimize the project: individual adjustment of the size of a transistor, to best meet the requirements of the specification (power consumption, voltage levels, switching thresholds, transient response, geometrical dimensions, etc.)
- Free to choose the most suitable topology (placement and routing) for each block.
- Free to choose the most suitable solutions to organize the relationships between neighboring blocks.
- Globally optimal choice of organization of supply chains, ground and synchronization.

Full Custom Design Approach

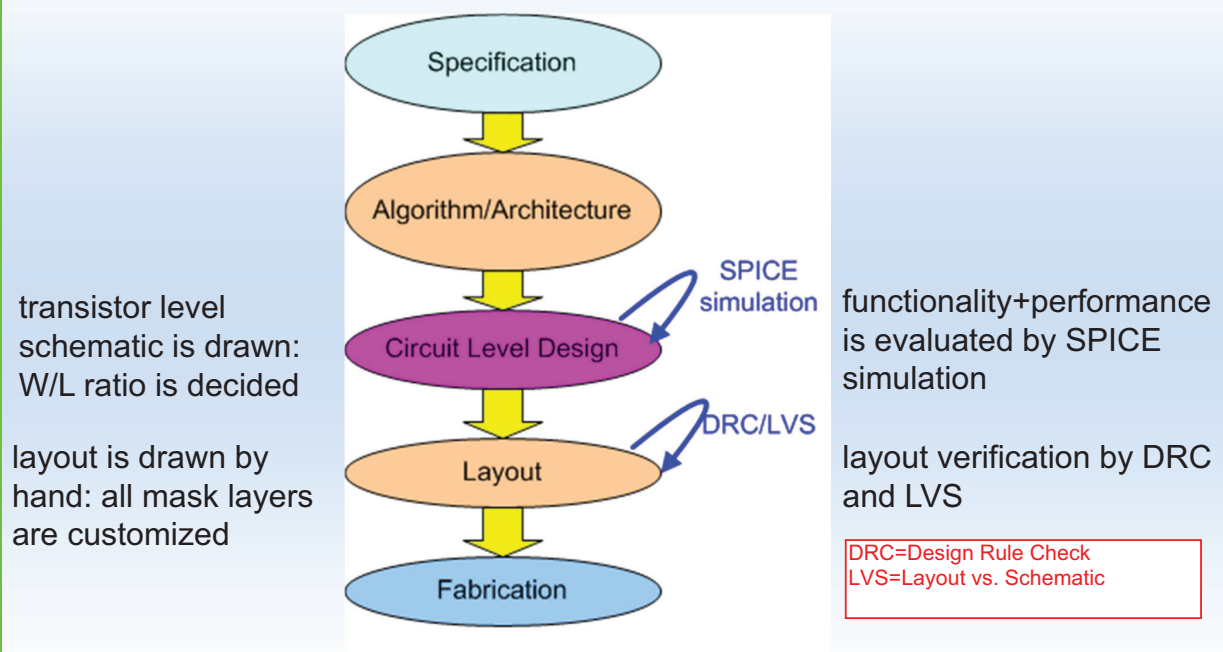
Advantages

- Full control over all aspects of the project
- Possibility of achieving the highest possible performance
- Possibility of achieving the lowest possible power consumption
- Minimum possible topological space of the crystal

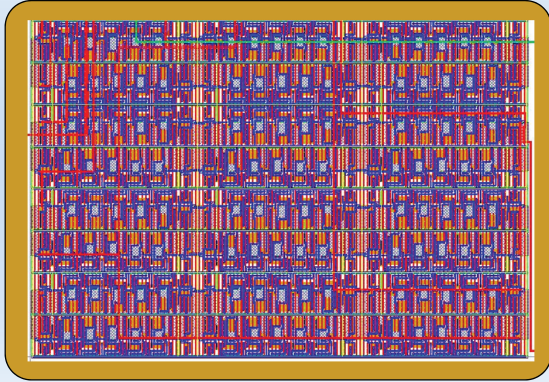
Disadvantages

- Long duration of the project
- High qualified IC design engineers needed

Full Custom Design Flow



Full Custom Layout



[Synopsys]

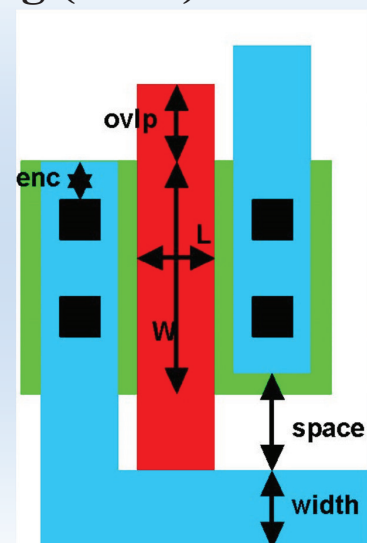
- Use of capabilities of all layers.
- Use of any geometry, topology, which is considered the best.
- Extraction of only the technological design rules.

Layout Verification

1. Design Rule Checking (DRC)

requirement for manufacturability

- DRC checker looks for errors
 - width
 - space
 - enclosure
 - overlap

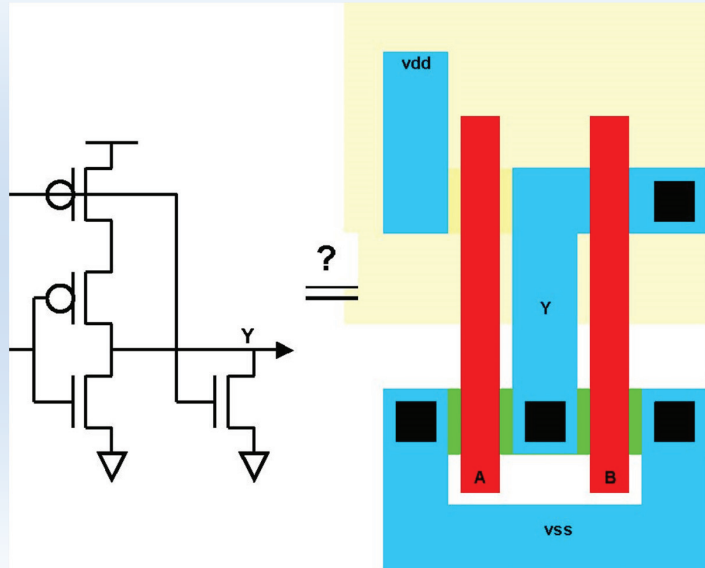


Layout Verification

2. Layout versus Schematic (LVS)

Extracts netlist from layout by analyzing polygon overlaps

Compares extracted netlist with original schematic

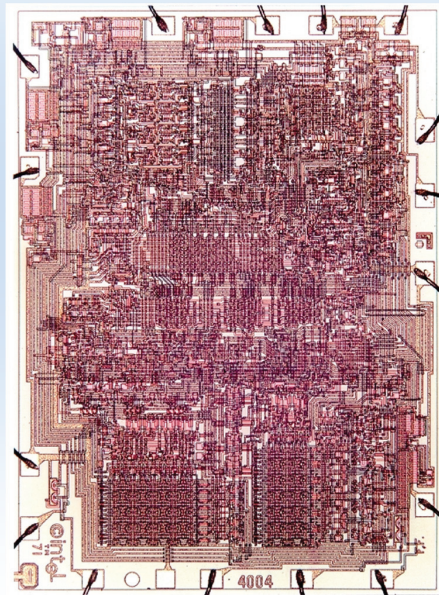


Reasons for Full Custom Design

- There are no suitable existing cell libraries available
- Existing logic cells are not fast enough
- Logic cells are not small enough
- Logic cells consume too much power
- VLSI is so specialized that some circuits must be custom designed
- Trends: fewer and fewer full custom ICs are being designed (mainly analog and mixed-signal ICs)

Practice of Full Custom Design (1)

Example (1971): μ P Design



Intel 4004

1000 transistors
1 MHz operation

courtesy: Intel

Prof. Schumann

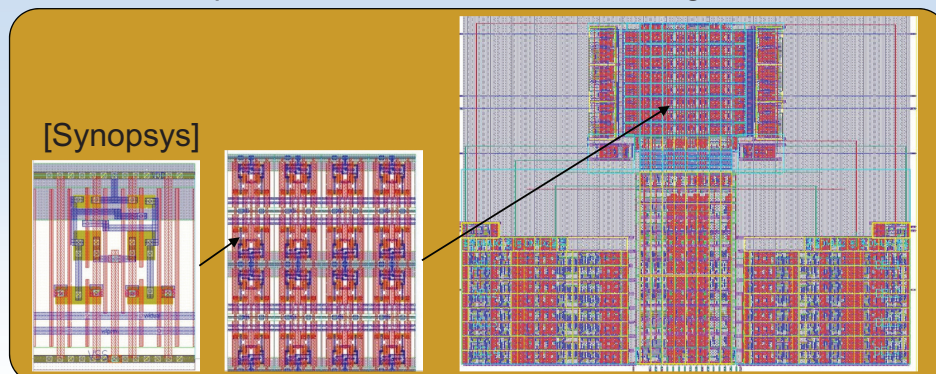
VLSI Design and Testing

169

Practice of Full Custom Design (2)

- The main objective of full custom design is to ensure fine-grained regularity and modularity.
- Layout-level modularity and regularity are essential for successful full custom design
- Regularity usually ensures a better area utilization.
- Dense layout results in lower parasitics.

Example: Full Custom SRAM Cell Design



Prof. Schumann

VLSI Design and Testing

170

Trend: Transition to Semi Custom Design



Microphotograph
of Pentium 4

Courtesy: Intel

Enhance Design Productivity: Cell-Based Design

The Intel® Pentium® 4 processor on 90nm technology is the first Intel microprocessor whose significant portion (~50% of the non-cache devices) was designed using a Cell-Based Design (CBD) methodology. In the CBD methodology, Electronic Design Automation (EDA) tools are used with a library of standard cells to build up a large and complex design.

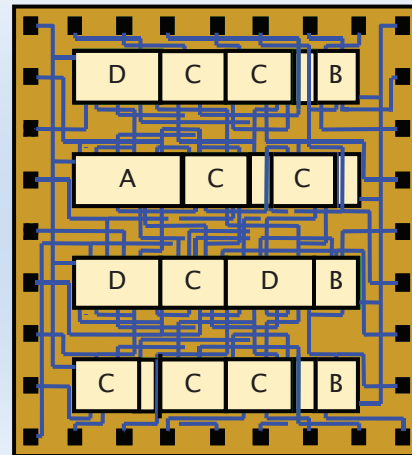
CBD usage in the Pentium® 4 processor

	CBD	Non-CBD
Area	52%	48%
Cell count	44%	56%
Transistor count	50%	50%

[Intel 2004]

Cell-Based (Standard Cell) Design Approach

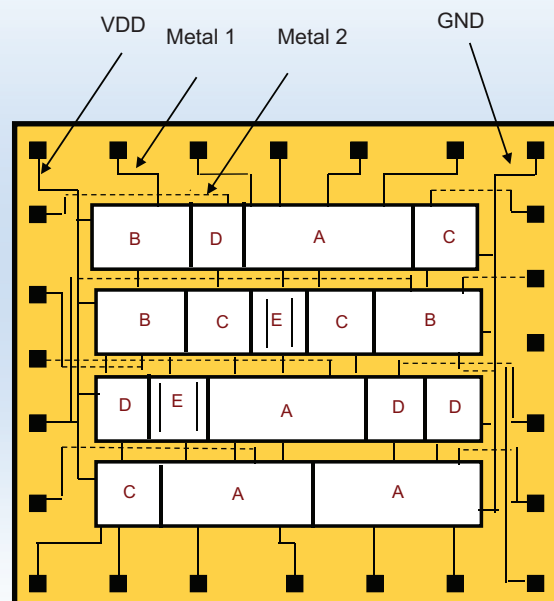
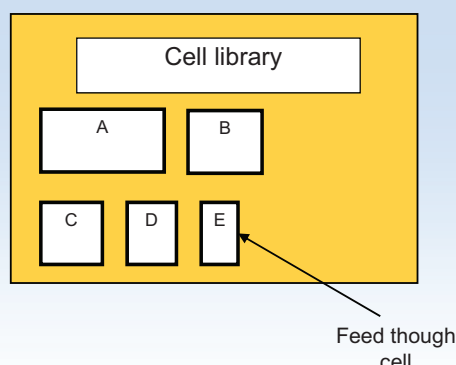
- Cell Library:
 - Cells are developed once.
 - Cells are highly optimized.
- Topology:
 - Cells are placed in a row (line).
 - Channels are used for interconnects.
 - Routing is performed on top of cells.
- Standard cells can be overlapped with macrocells (such as RAM).



[Synopsys]

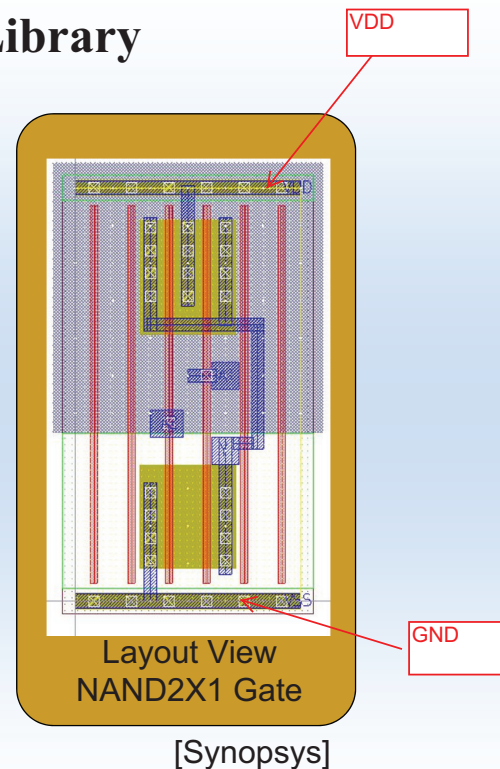
Standard Cell Design

IC designer uses cells from the library. Design entry is done by schematic or a hardware description language (HDL). Layout is automatically placed and routed by EDA tools .



Standard Cell Library

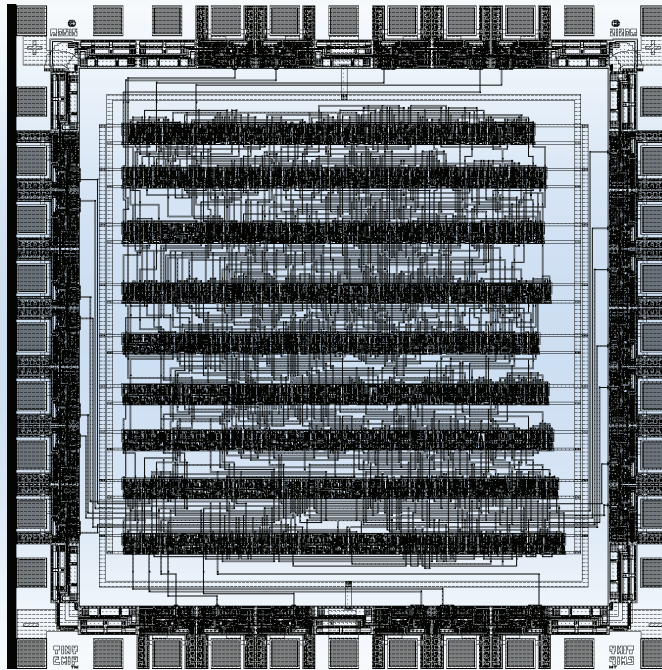
- A standard cell is a predesigned layout of one specific basic logic gate.
- Each cell usually has the same standard height.
- A standard cell library contains a varied collection of standard cells.
- Libraries are usually supplied by a VLSI vendor or a library group.



Standard Cell Library

- The main macro-parameters of digital circuits - speed, area (density units) and power consumption.
- General use, in which the three parameters are equally important and the library is built on the basis of trade-offs between these three parameters
- High-speed, which focuses on getting the most possible performance within the capabilities of the technology used
- With low power consumption, in which the privileged condition is to achieve low power consumption.

Practice of Standard Cell Design



[fig. from: Rabaey et al, Digital Integrated Circuits, 2nd Edition]

Prof. Schumann

VLSI Design and Testing

177

New Generation Standard Cell Design



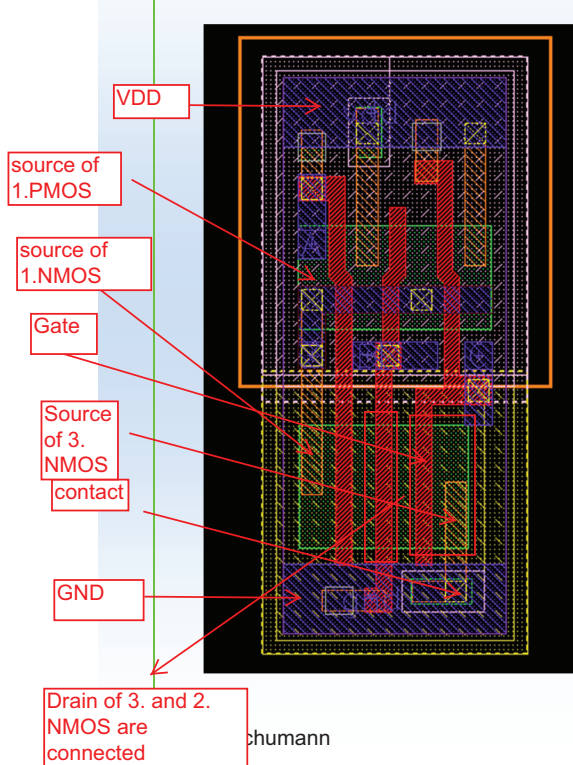
[fig. from: Rabaey et al, Digital Integrated Circuits, 2nd Edition]

Prof. Schumann

VLSI Design and Testing

178

Standard Cell Example (2)



VDD=1.5V (nominal supply voltage)

worst case scenario: lowest supply voltage, highest temp

best case scenario

Path	1.2V - 125°C	1.6V - 40°C
$In1-t_{pLH}$	$0.073+7.98C+0.317T$	$0.020+2.73C+0.253T$
$In1-t_{pHL}$	$0.069+8.43C+0.364T$	$0.018+2.14C+0.292T$
$In2-t_{pLH}$	$0.101+7.97C+0.318T$	$0.026+2.38C+0.255T$
$In2-t_{pHL}$	$0.097+8.42C+0.325T$	$0.023+2.14C+0.269T$
$In3-t_{pLH}$	$0.120+8.00C+0.318T$	$0.031+2.37C+0.258T$
$In3-t_{pHL}$	$0.110+8.41C+0.280T$	$0.027+2.15C+0.223T$

3-input NAND cell
(from ST Microelectronics):

C = Load capacitance
T = input rise/fall time

3 NMOS in series
3 PMOS in parallel

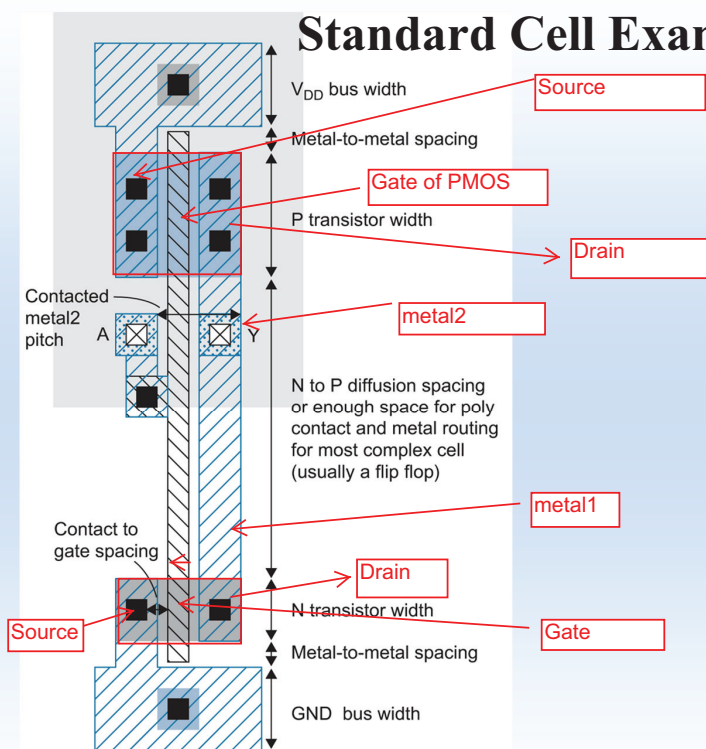
[fig. from: Rabaey et al, Digital Integrated Circuits, 2nd Edition]

Prof. Schumann

VLSI Design and Testing

179

Standard Cell Example (1)



CMOS-Inverter: 1 NMOS + 1 PMOS

FIG 8.29 Typical standard cell layout with some [1] of the constraints

Prof. Schumann

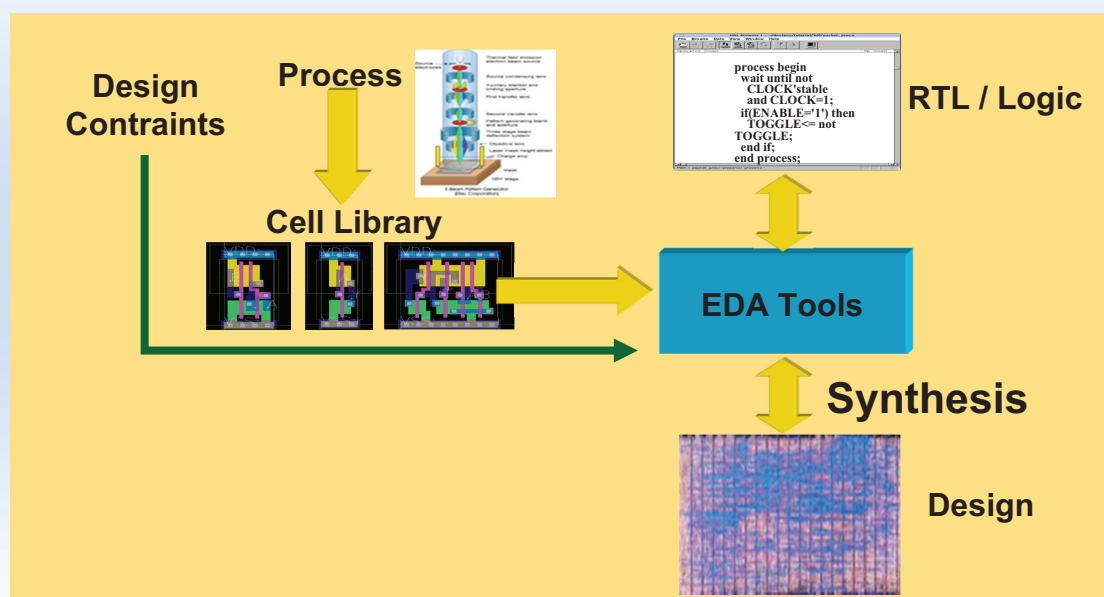
VLSI Design and Testing

180

Features of Standard Cells

- Standard cell - a logical unit that performs a specific function such as NAND, XOR, NOR, etc.
- The library of cells - the data on standard cells (function, structure, structure of outputs, topology, technology, etc.)
- The cells have the same height and provide: floorplan and layout of topological design, simplified placement and flow of interconnects.
- Placing the cells in rows provides application of more simple methods of channel routing.

Standard Cell Design: Automatic Synthesis



Generalized Standard Cell Design Flow

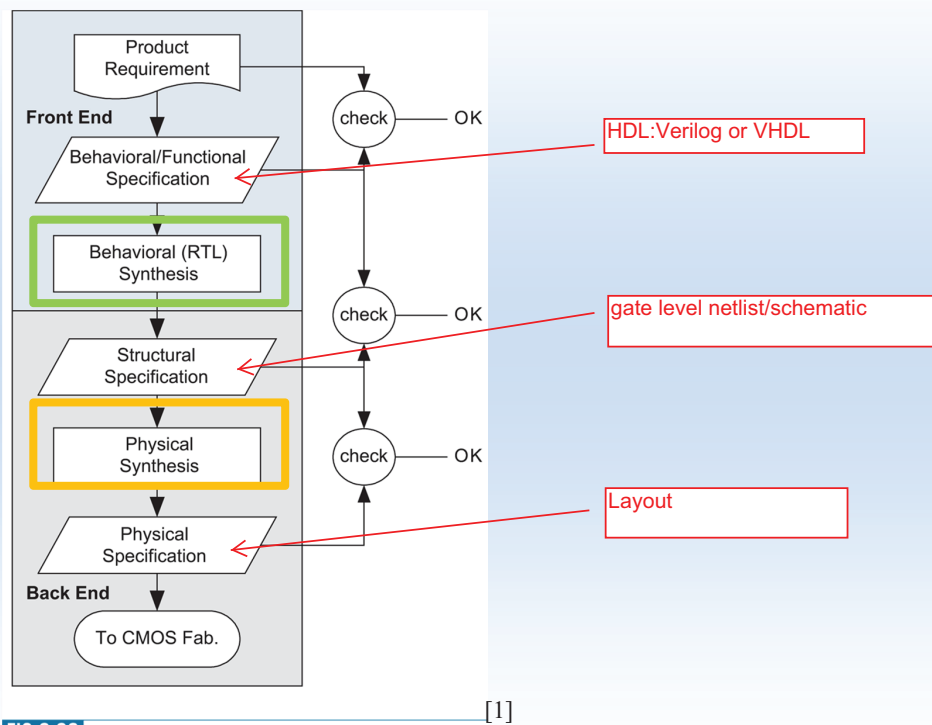


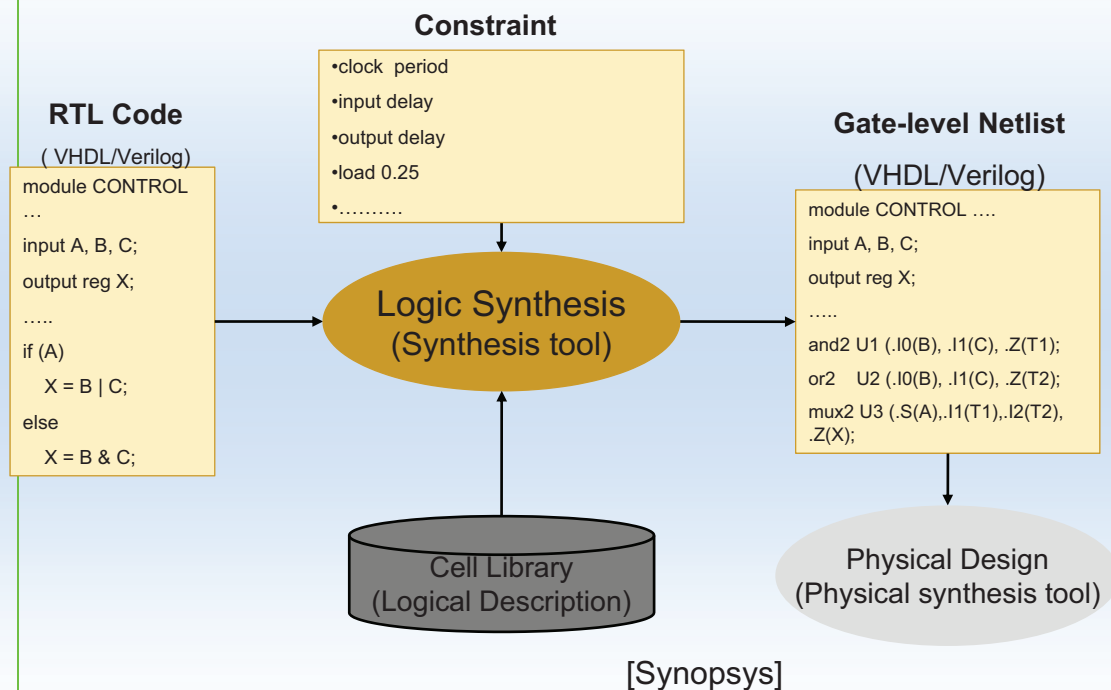
FIG 8.38 Generalized design flow

Prof. Schumann

VLSI Design and Testing

183

Standard Cell Design Flow: RTL Synthesis

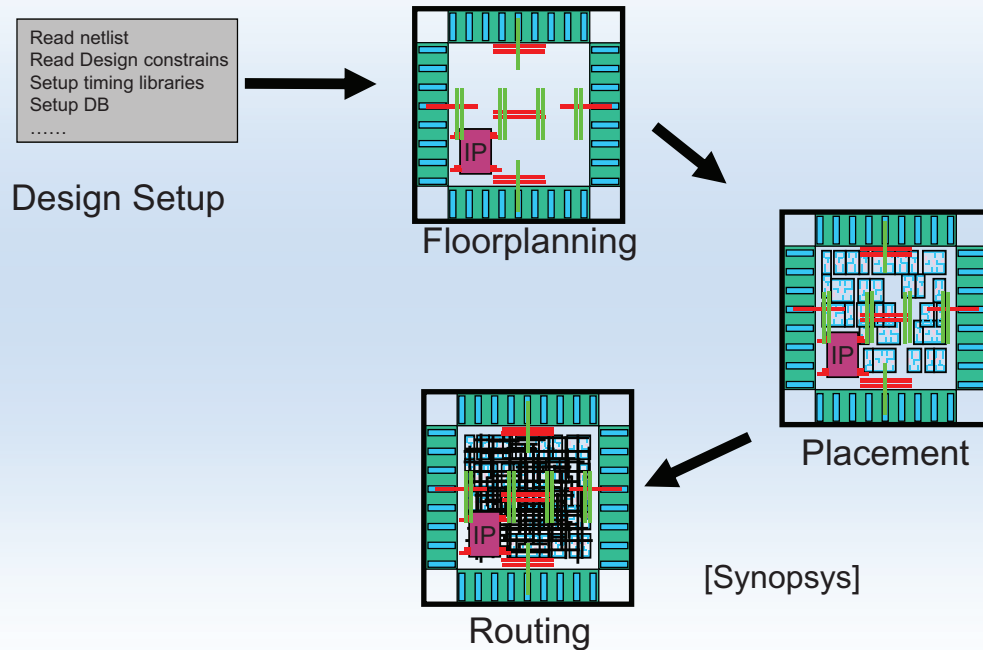


Prof. Schumann

VLSI Design and Testing

184

Standard Cell Design Flow: Physical Synthesis



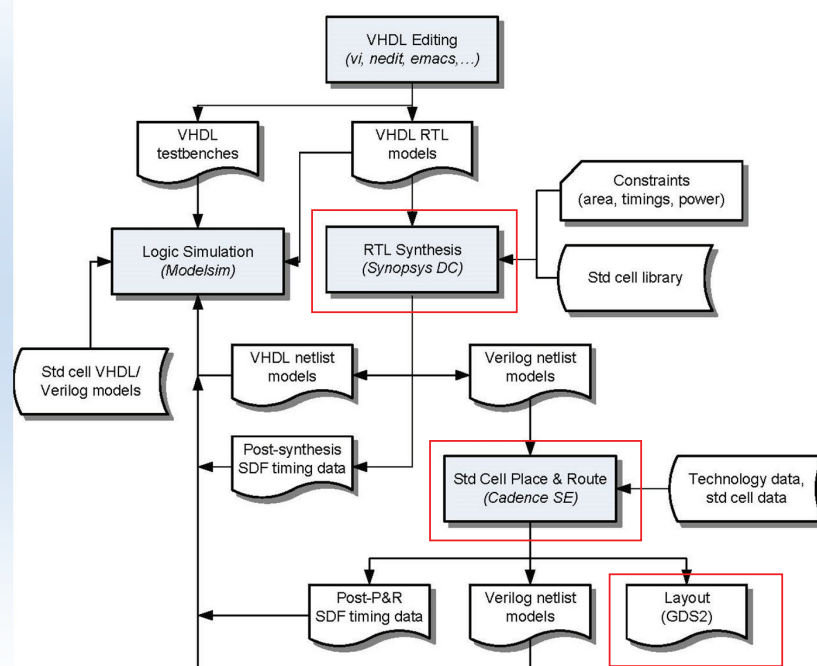
Prof. Schumann

VLSI Design and Testing

185

Standard Cell Design Flow: EDA Tools

Figure 1.1: Top-down design flow.



[A.Vachoux:Top-down digital design flow]

Prof. Schumann

VLSI Design and Testing

186

RTL (Behavioral) Synthesis: Design Compiler will select the appropriate architecture for your constraints; targeted logic gates belongs to a library provided by a foundry or IP company

Physical Synthesis (standard cell place & route): geometric realization of the gate-level netlist is done, a so-called *layout* generated; logic cells are put together in rows of equal heights; each cell has a power rail at its top and a ground rail at its bottom.

Detailed RTL (Behavioral) Synthesis Flow

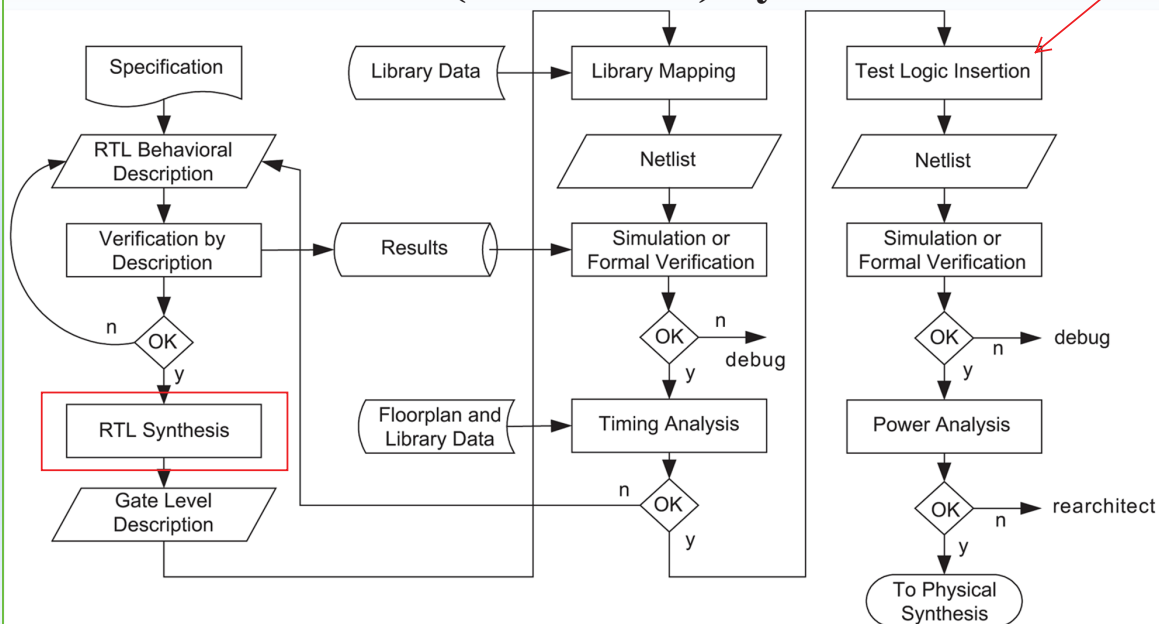


FIG 8.39 RTL synthesis flow

[1]

Detailed Physical Synthesis Flow

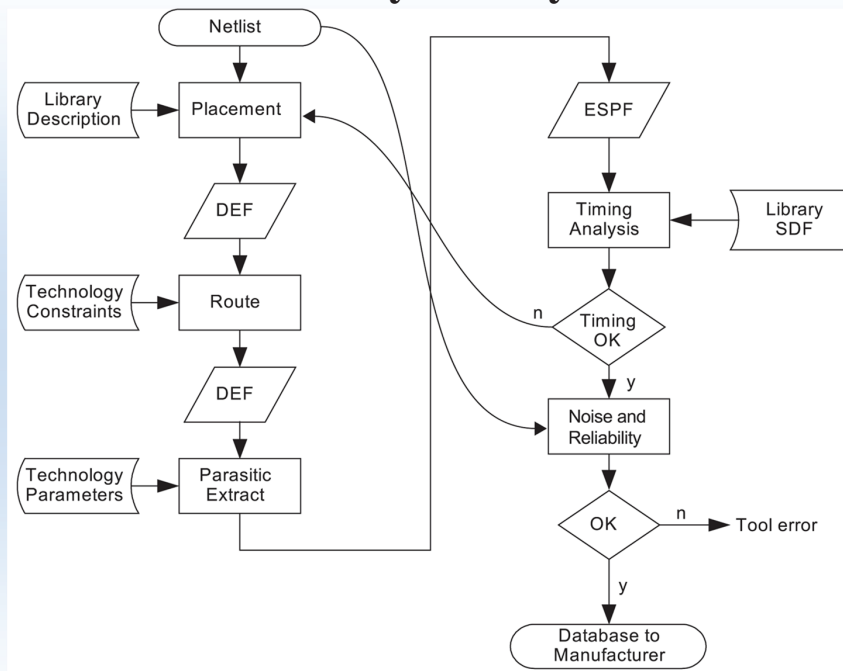


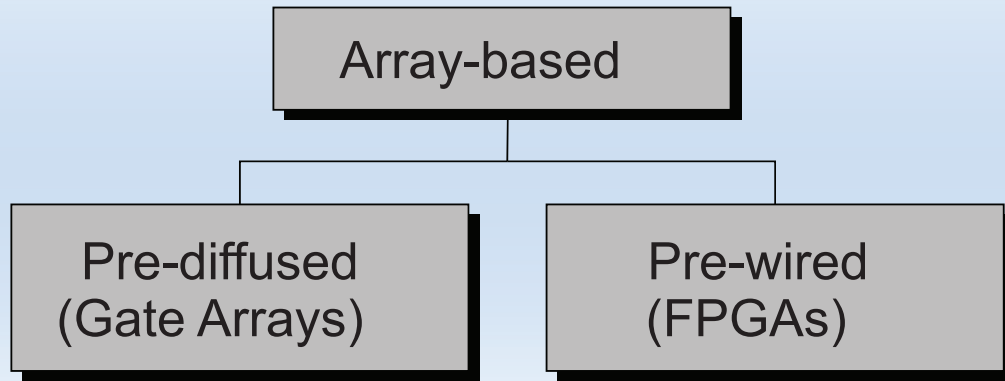
FIG 8.41 Standard cell place and route design flow

[1]

Fabless IC Design Company Model

- Company does design only. Fabrication is performed by another company, a so called foundry (e.g. TSMC, UMC).
- Physical Synthesis (place and route, etc.) might be performed at that company or with their assistance
- Top 5 fabless IC companies (2019): Broadcom, Qualcomm, Nvidia, MediaTek, AMD
- Foundries became the cornerstone of the fabless model – providing a manufacturing partner for fabless IC companies.

Array-Based Implementation Approaches



[fig. from: Rabaey et al, Digital Integrated Circuits, 2nd Edition]

Array-Based Implementation Approaches

All of the cell-based implementation approaches require a complete run through the fabrication process -> cost of process run, takes 2 to 6 month for processing

Array-based implementation approaches do not require a complete run through the manufacturing process or they avoid dedicated processing completely!

Advantage: lower cost for small series
 faster time-to-market
 possibility of late changes in the design

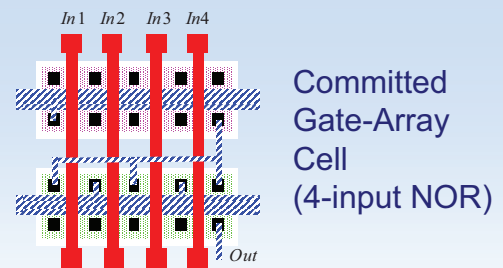
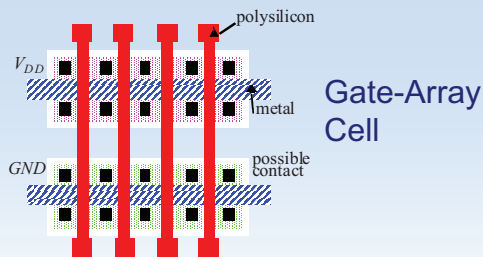
Disadvantage: lower performance
 lower integration density
 higher power dissipation

I. Gate Arrays

Def. Gate Arrays: Pre-manufactured wafers containing arrays of transistors

To transform the wafers into an actual design, only the interconnections between the transistors have to be added. This is done by only a few metallization steps! The metal layers are designed for the actual circuit and are applied to the pre-manufactured wafer!

Advantage: fabrication of metal layers takes a week or even less!



[fig. from: Rabaey et al, Digital Integrated Circuits, 2nd Edition]

II. FPGAs

FPGA (Field Programmable Gate Array):

- Chips are prefabricated with logic blocks and interconnects.
- Interconnects are predefined wire segments of fixed lengths with switches in between.
- Logic and interconnects can be programmed (erased and re-programmed) by users. No fabrication is needed.

FPGAs

Advantage: manufacturing process is completely separated from the design phase. This means a very short turnaround time!

A simple comparison

Technology	NRE** cost*	Time to get hardware*
ASIC (cell-based)	\$20K-\$100K	months
FPGA	\$100-\$100K	minutes/hours

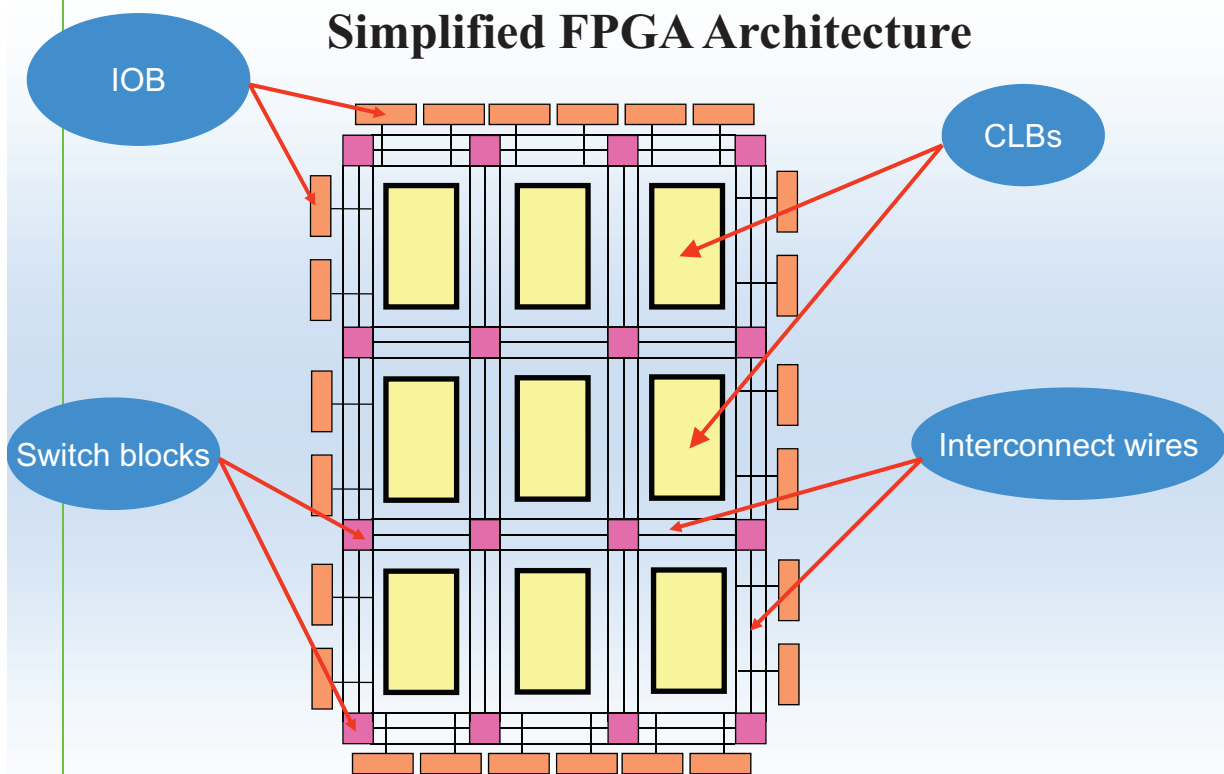
* approximate values to give an idea

** Non-recurring engineering (NRE) refers to one-time cost of specifying and designing a new IC

Applications of FPGAs

- High speed serial interconnections
- Memory controllers and interfaces
- Digital Signal Processing
- Video Processing
- ASIC prototyping

Simplified FPGA Architecture



Prof. Schumann

VLSI Design and Testing

197

Basic Blocks on FPGA

CLB

This is a acronym for Complex Logic Block

Interconnect wires

These internal wires are used to carry signal between different CLBs

Switch blocks

Switch blocks are useful to connect/disconnect wires

IOB

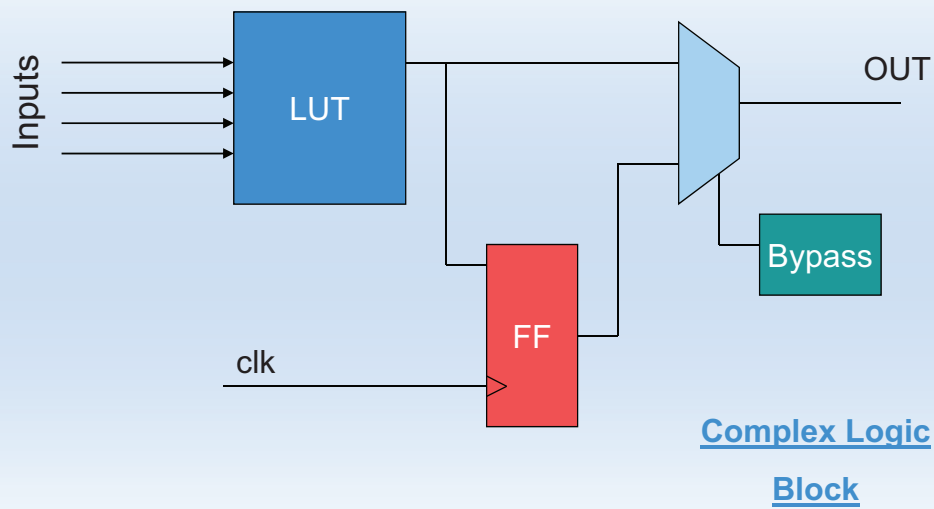
These input and output blocks are used to configure the pins as input or output and also to register the signals

Prof. Schumann

VLSI Design and Testing

198

Simplified CLB



Prof. Schumann

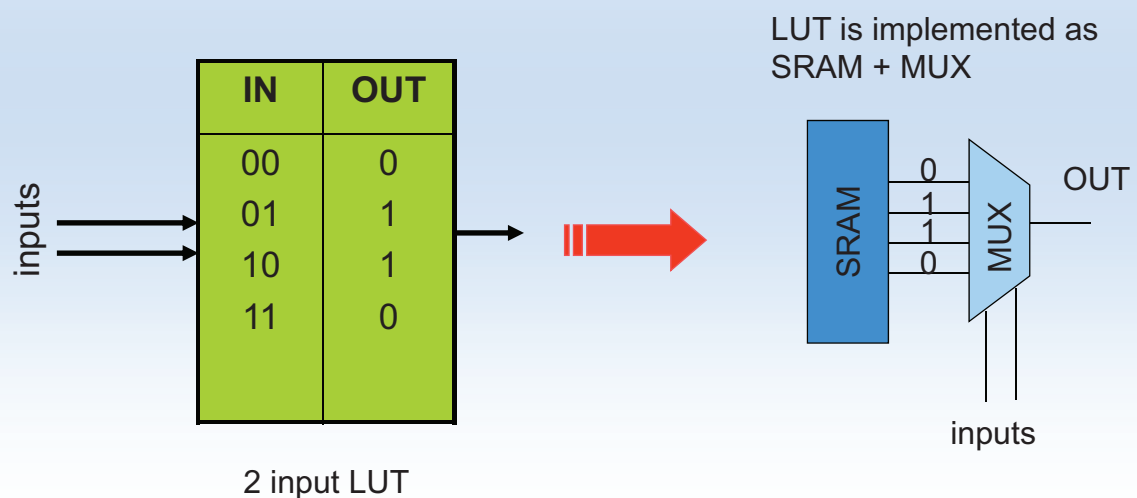
VLSI Design and Testing

199

Look-Up Table (LUT)

How to implement „programmable“ logic, logic that can perform any possible Boolean function?

A lookup table (LUT) is programmed to capture the truth table of a specific function



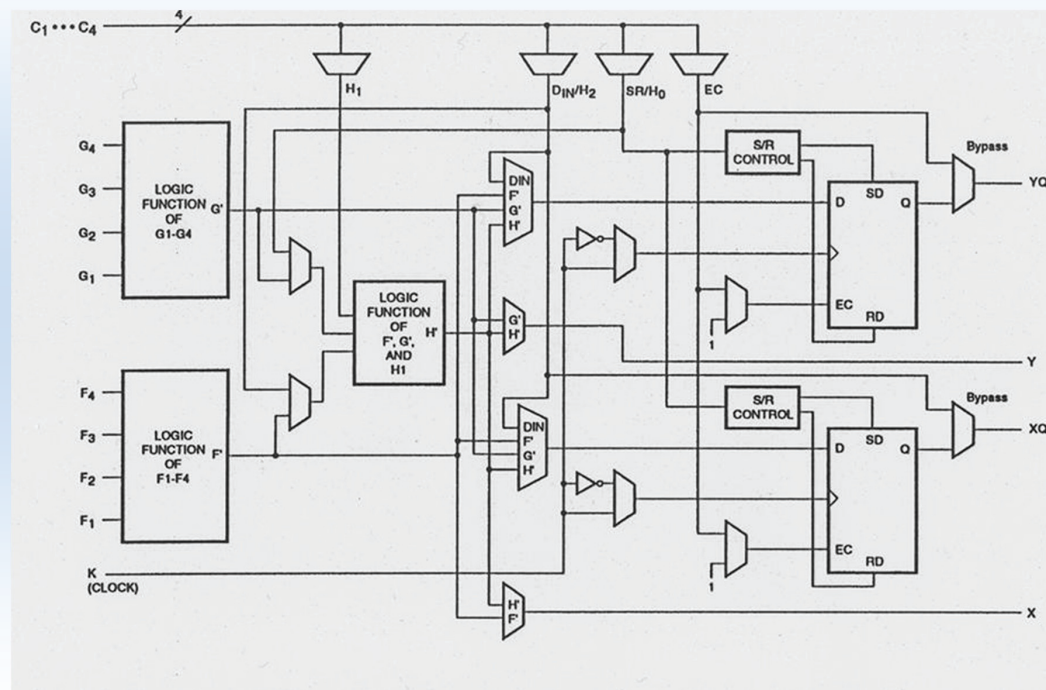
Prof. Schumann

VLSI Design and Testing

200

High Performance CLB

Xilinx XC4000 CLB



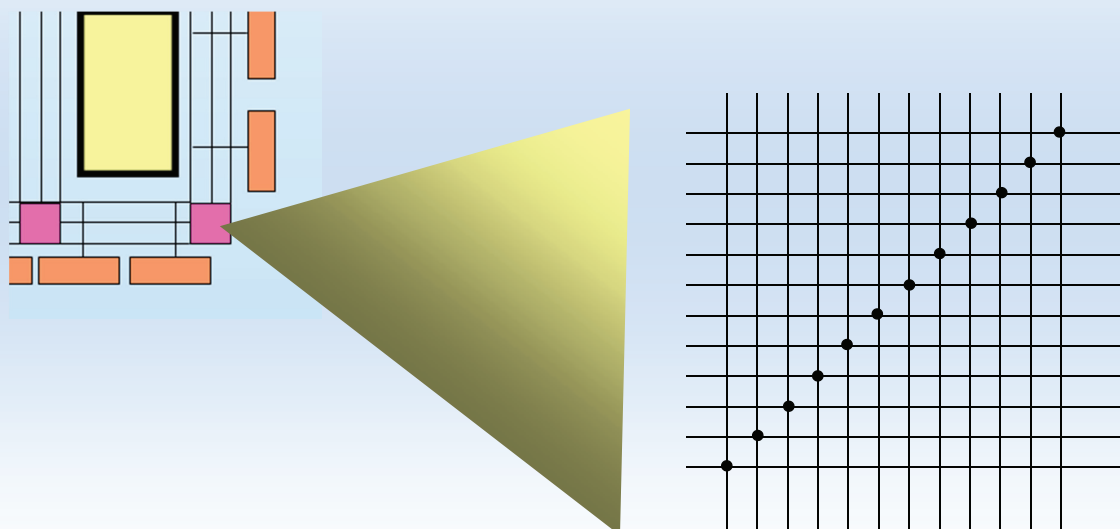
Prof. Schumann

VLSI Design and Testing

201

Switch Blocks on FPGA

Programmable Switching Block



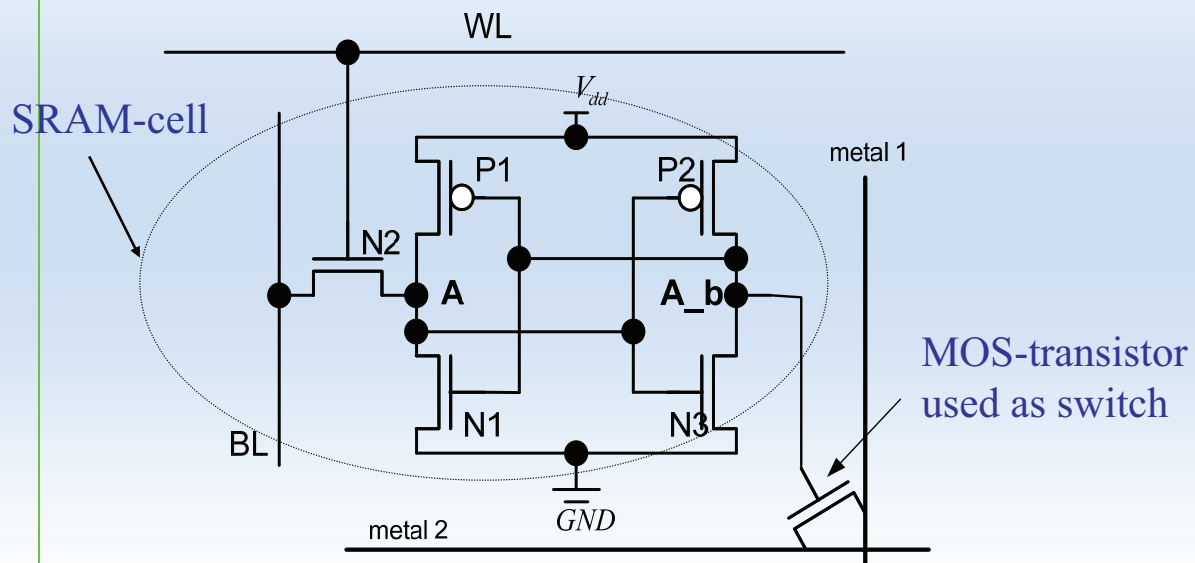
Prof. Schumann

VLSI Design and Testing

202

Programmable Switch on FPGA

SRAM-based Programmable Switch



Prof. Schumann

VLSI Design and Testing

203

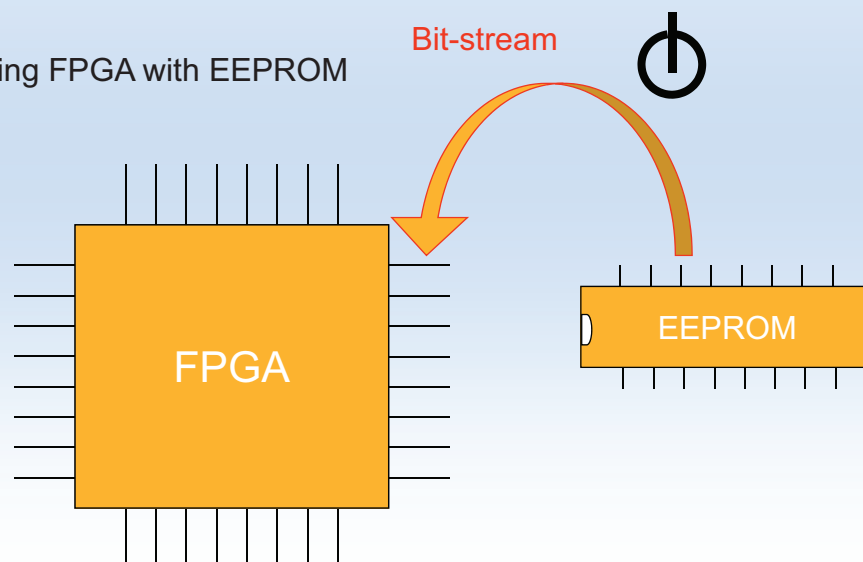
FPGA Programming

Most popular approach to implement LUTs: **SRAM**

Advantage: FPGA can be fabricated using standard CMOS process technology

Disadvantage: Configuration is lost when power supply is switched off -> For reconfiguration an external EEPROM is used

Configuring FPGA with EEPROM



Prof. Schumann

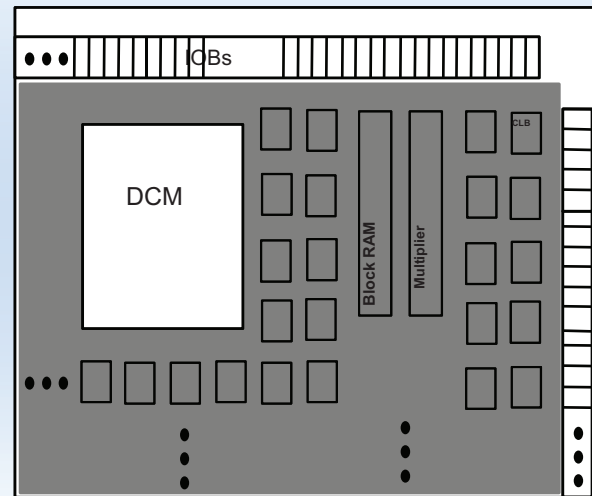
VLSI Design and Testing

204

High-Performance FPGAs

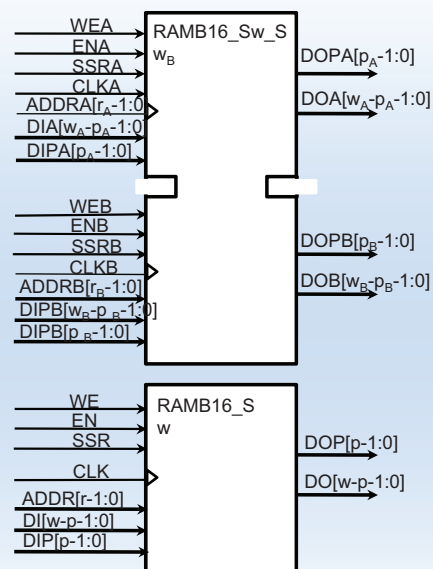
Key components

- Block RAM
- Multiplier Blocks
- Digital Clock Manager (DCM) Blocks



Block RAM

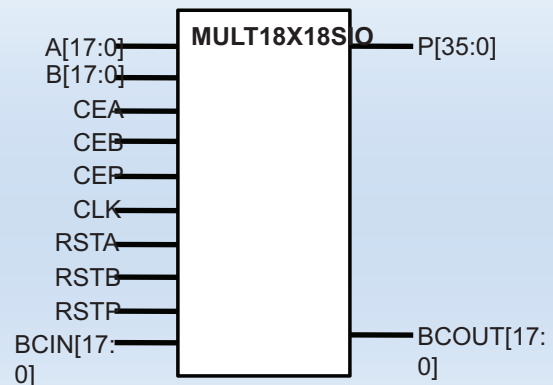
For applications requiring large, on-chip memories, FPGAs provide efficient RAM memory blocks. Using various configuration options, Block RAM create FIFOs, large look-up tables, data width converters, circular buffers, and shift registers, each supporting various data widths and depths.



Multiplier Blocks

18x18 multipliers speed up DSP logic in the FPGA. The multipliers are fast and efficient at implementing signed or unsigned multiplication of up to 18 bits.

The multipliers can be cascaded with each other or with CLB logic for larger or more complex functions.



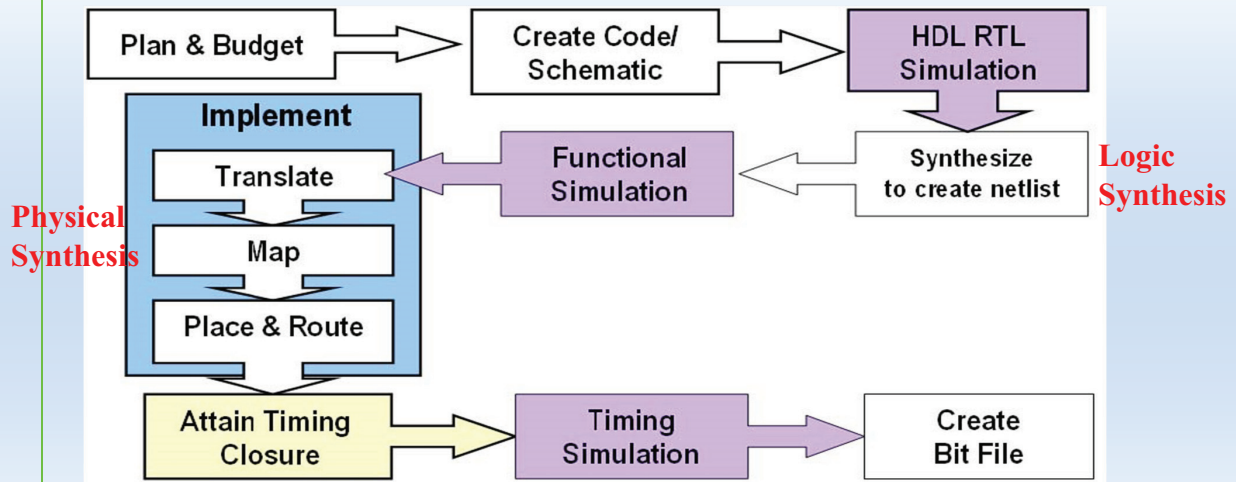
Digital Clock Manager (DCM)

Digital Clock Managers (DCMs) integrate advanced clocking capabilities directly into the FPGA's global clock distribution network. Consequently, DCMs solve a variety of common clocking issues, especially in high-performance, high-frequency applications:

- Phase Shift a clock signal, either by a fixed fraction of a clock period or by incremental amounts.
- Multiply or Divide the clock frequency of an incoming clock signal.
- Condition a Clock, ensuring a clean output clock with a 50% duty cycle.

Feature	Description	DCM Signals
Frequency Synthesizer Output	Multiply CLKIN by the fraction (M/D) where $M = \{2..32\}$, $D = \{1..32\}$	CLKFX CLKFX180
Clock Divider Output	Divide CLKIN by 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 9, 10, 11, 12, 13, 14, 15, or 16	CLKDV

FPGA Design Flow



Prof. Schumann

VLSI Design and Testing

209

FPGA Design Flow

- The FPGA software major task, in addition to facilitate design-entry, is to synthesize and place-and-route the design.
 - Synthesis takes the design (HDL or schematic) and creates a flat netlist out of it.
 - Translate/Map/Place&Route describes several processes where the netlist elements are physically places and mapped to the FPGA physical resources.
 - After synthesis and place-and-route, there is a binary file that is ready to be "downloaded" into the FPGA.

Prof. Schumann

VLSI Design and Testing

210

Trade-Off ASIC vs. FPGA

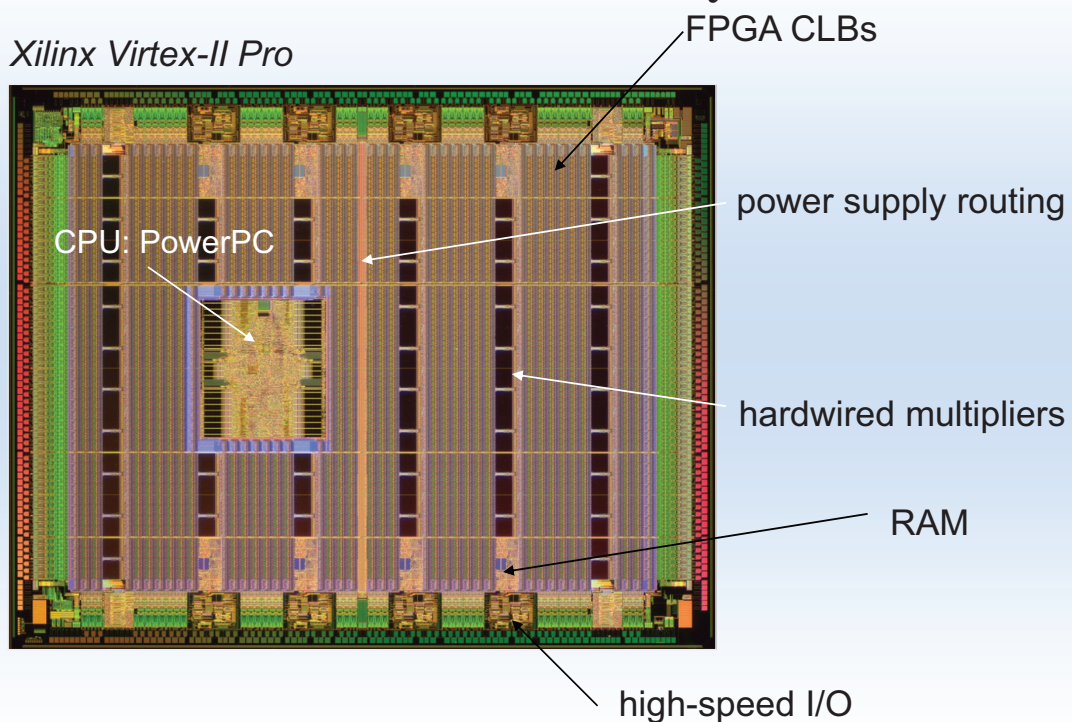
Trade-off Summary



Performance	NRE	Unit cost	TTM
ASIC	ASIC	FPGA	FPGA
FPGA	FPGA	ASIC	ASIC

FPGA for Embedded Systems

Xilinx Virtex-II Pro



System-on-a-Chip Design (SoC)

- **Characteristics**
 - Very large transistor counts on a single IC
 - Mixed technologies on the same chip
 - Digital, memory, analog,
 - FPGA
 - Hardware and software
 - Multiple clock frequencies
 - Hierarchical design with embedded reusable IP cores

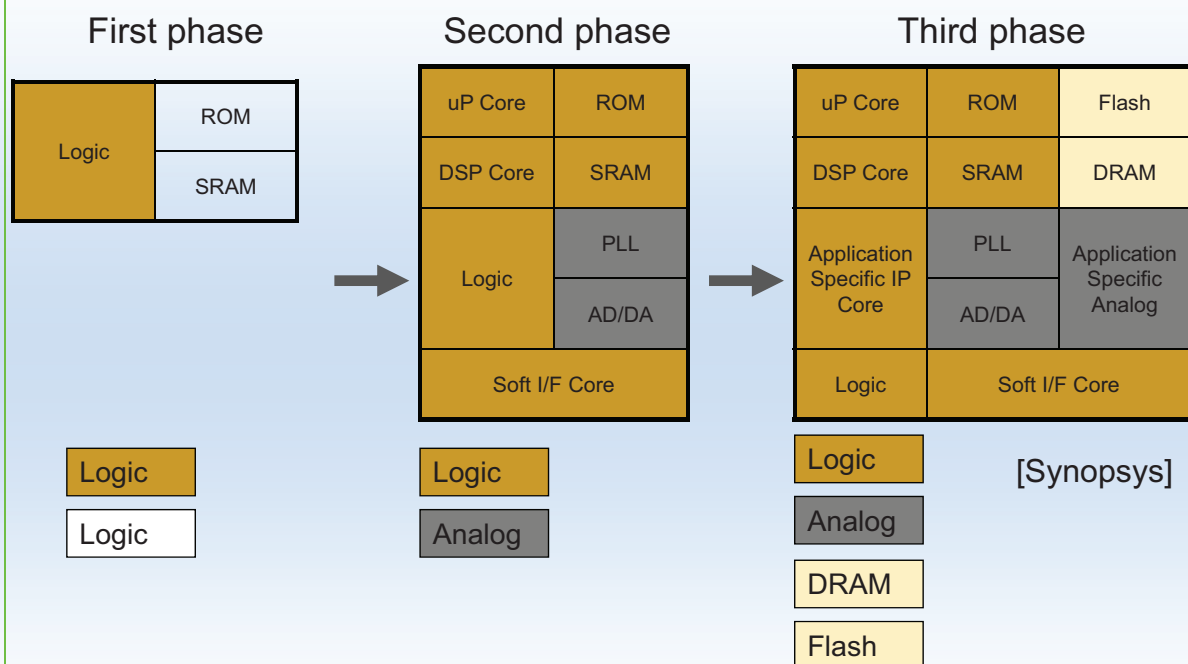
System-on-a-Chip Design (SoC)

- **Reasons**
 - Complex applications
 - Semiconductor density increases 58% per year, but design productivity increases 21% annually.
 - Process technology allows it
 - High performance
 - Miniaturization
 - Battery life
 - Short market windows
 - Cost sensitivity

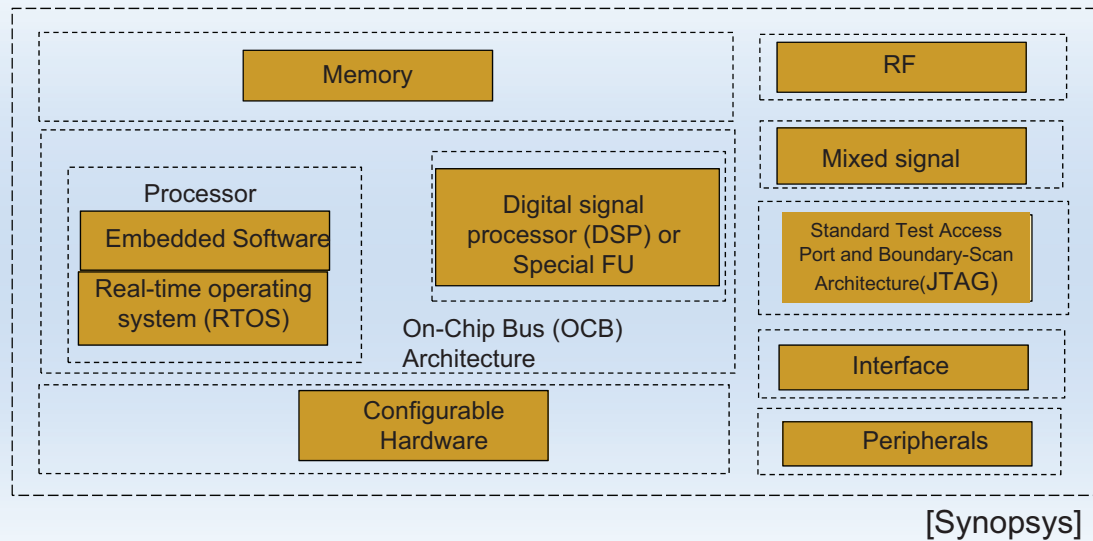
System-on-a-Chip Design (SoC)

- **Reasons**
 - Complex applications
 - Semiconductor density increases 58% per year, but design productivity increases 21% annually.
 - Process technology allows it
 - High performance
 - Miniaturization
 - Battery life
 - Short market windows
 - Cost sensitivity

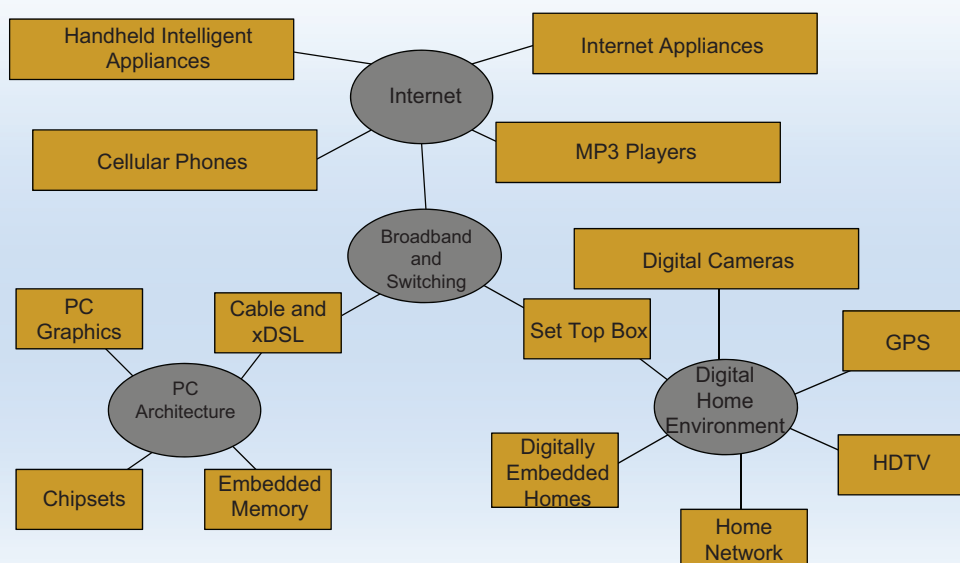
SoC Evolution



SoC Architecture



SoC Applications



Source: Semico Research Corporation

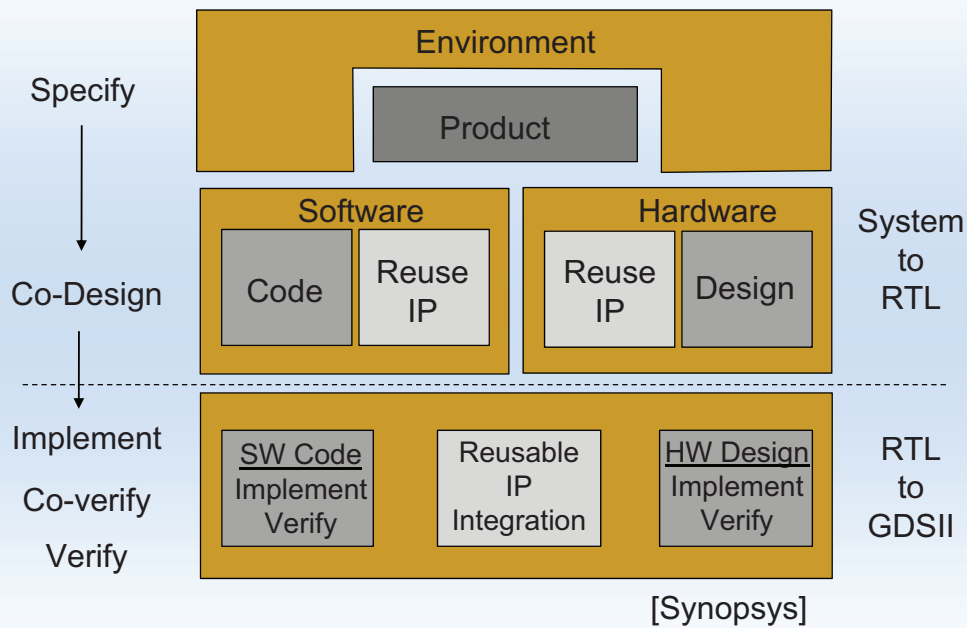
Features of SoC Design

- Core based design approach
 - Design Complexity
 - Time to Market
 - Core : A pre-designed, pre-verified silicon circuit block. E.g. Microprocessor, Bus Interface, BIST Logic, SRAM, Memory.
- Core integration
 - Re-usable cores: different types, different vendors
 - User defined logic

Features of SoC Design

- Designing cores for integration
 - Parameterization
 - Customizable soft cores. Core provider supplies essential set of pre-verified parameters
 - Functionality
 - Single core - preferable
 - Multiple core - needs good partitioning
 - Interface
 - Support standard buses to ease integration

Organization of SoC Design



SoC Cores

- Cell/Macro Library elements
- DSPs, Microcontrollers, FPGAs
- Implementation of standards
 - Function (MPEG, JPEG, CRC, PicoJava, etc.)
 - Interconnects (PCI, SCSI, USB, 1394, IrDA, Bus Bridges)
 - Networking (10/100 Ethernet, ATM, etc.)

SoC Core Types

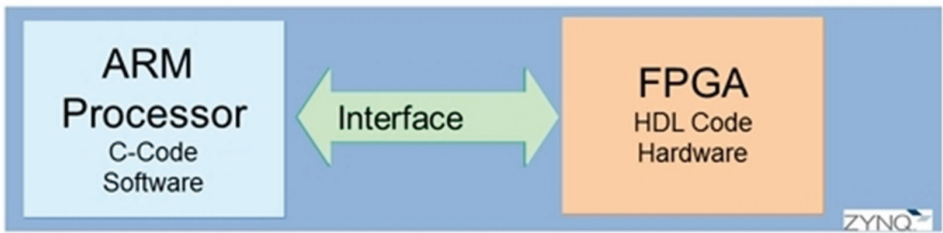
CMOS-technology

- **Soft Cores:** Technology independent synthesizable description. White box implementation – visible and modifiable. Core can be extended functionally.
- **Firm Cores:** Technology dependent gate level netlist. Internal implementation of core cannot be modified. User can parameterize I/O to remove unwanted functionality.
- **Hard Cores:** Layout and timing information provided. Cannot be re-synthesized. Integration is simple and can result in highly predictable performance.

SoC Device: Zynq

MathWorks

What is Zynq?



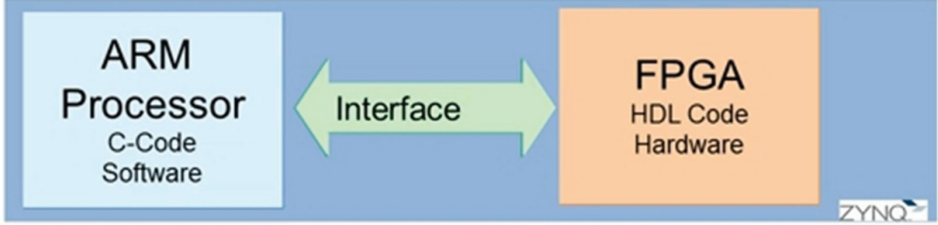
- New product family from Xilinx®
 - All Programmable System on Chip (SoC)
- FPGA + ARM® on one chip
 - Enable high-performance system development
 - Reduces BOM cost over multi-chip solutions

ZYNQ

SoC Device: Zynq

MathWorks

Zynq Design Challenges



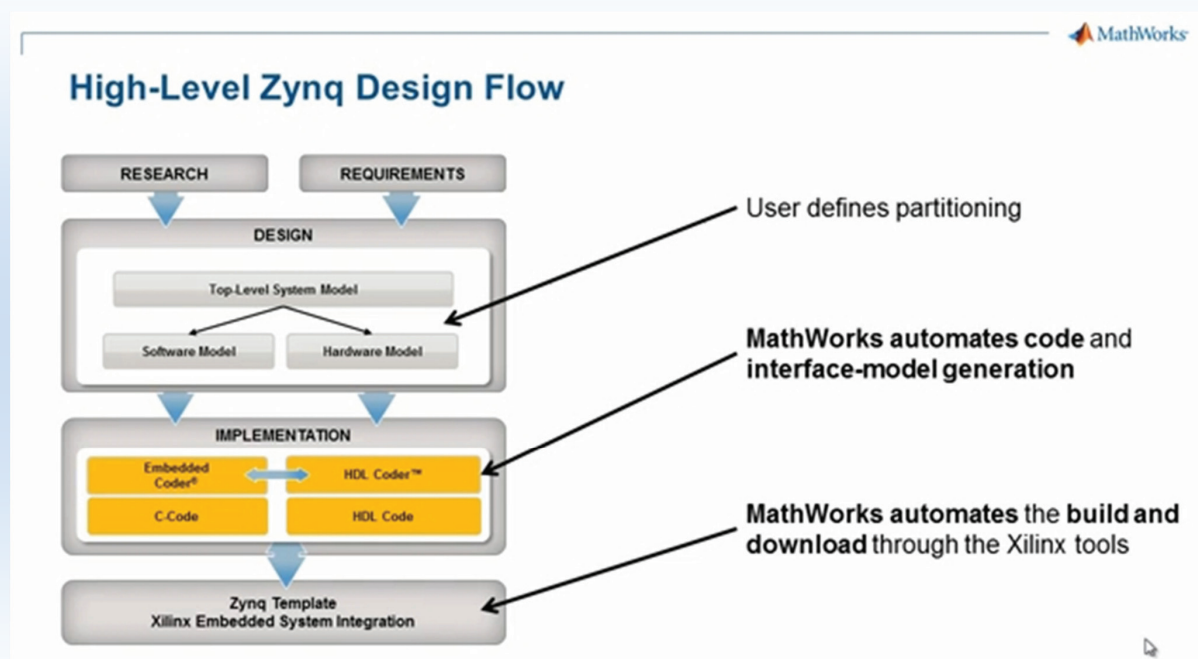
- **FPGA Designers** not familiar with programming processors
- **DSP/Processor programmers** not familiar with FPGAs
- **What should run on the FPGA vs. what should run on the ARM?**
- **No established rules for hooking up the interface** between FPGA and ARM processor

Prof. Schumann

VLSI Design and Testing

225

SoC Model-Based Design Flow

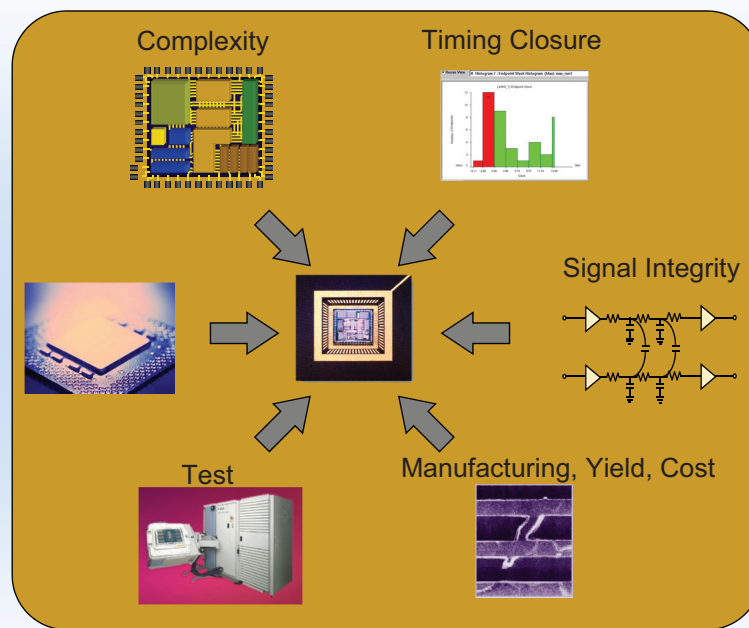


Prof. Schumann

VLSI Design and Testing

226

SoC Design Challenges



Source: <http://www.keyasic.com/>

Summary

To match this product need, the capability of a silicon CMOS chip will continue doubling every 2-3 years.

ASIC styles include full custom (for analog, memory, μP) and semi custom design: Cell-based and FPGA implementation.

New possibilities with SoC Design.